



Building RAG Agents with LLMs

Recent RAG Research:

From Retrieval Pipelines to Learned Search Agents

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본 자료의 저작권은 Jae Y. CHOI (jaeyeong2022@gmail.com)에게 있습니다.

본 자료는 자유롭게 이용할 수 있으나, 사용 시 저작자 및 출처를 반드시 표기해 주시기 바랍니다.

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Learning Objectives

From RAG Pipelines to Search Agents

- Explain search as a learned action
- Compare outcome and process feedback
- Distinguish search from deep web research
- Read results in their experimental context



From Pipeline to Policy

Who decides when and what to retrieve?

**The agent learns
when to search
and when to stop.**

**Retrieval: A tool returns external
evidence for the chosen query.**

**Augmented: New evidence
updates the context for the
next decision.**

**Generation: The model may
search again—or produce the
final answer.**

**User Asks
Something**

**Agent
Responds**

Three Research Questions

A map of five selected 2025 papers

What can the agent
do?

Actions

Search · inspect · draft

What gets rewarded?

Feedback

Outcome · process

Where does it learn?

Environment

Fixed corpus · open web

The retrieval pipeline becomes a sequential decision problem.



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Search Becomes an Action

The Shared Agent Loop

Reason → search → observe → reason → answer

Where was the author of The Hobbit born?

Reason

Find the author

<think>



Search

**The Hobbit
author**

<search>



Observe

J. R. R. Tolkien

Tool result



Continue

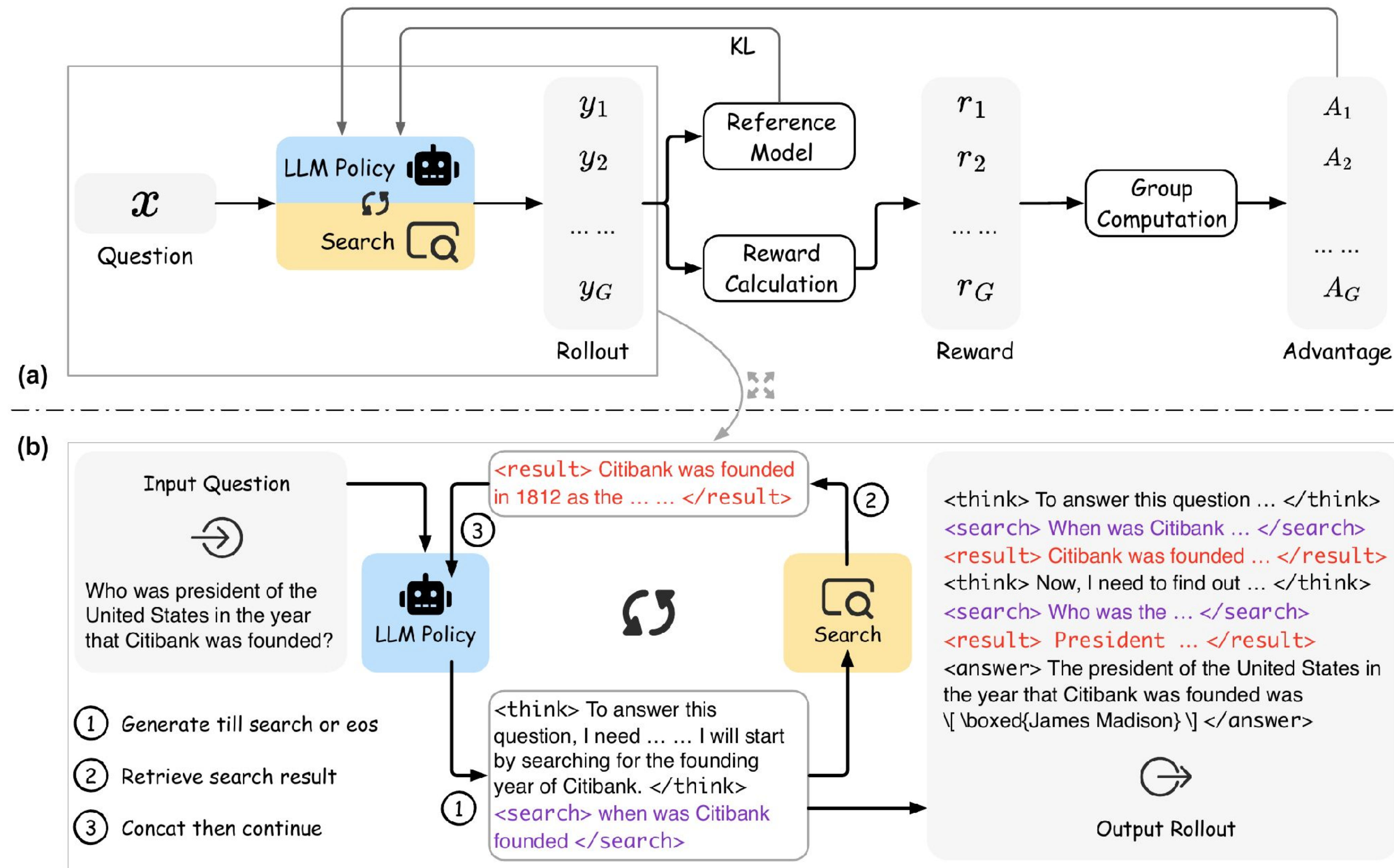
**Search
birthplace**

Then answer

$$a_t \sim \pi_{\theta}(\cdot \mid q, a_{<t}, o_{<t})$$

ReSearch

Learn reasoning with search—without annotated reasoning traces



GRPO

Answer F1 + format

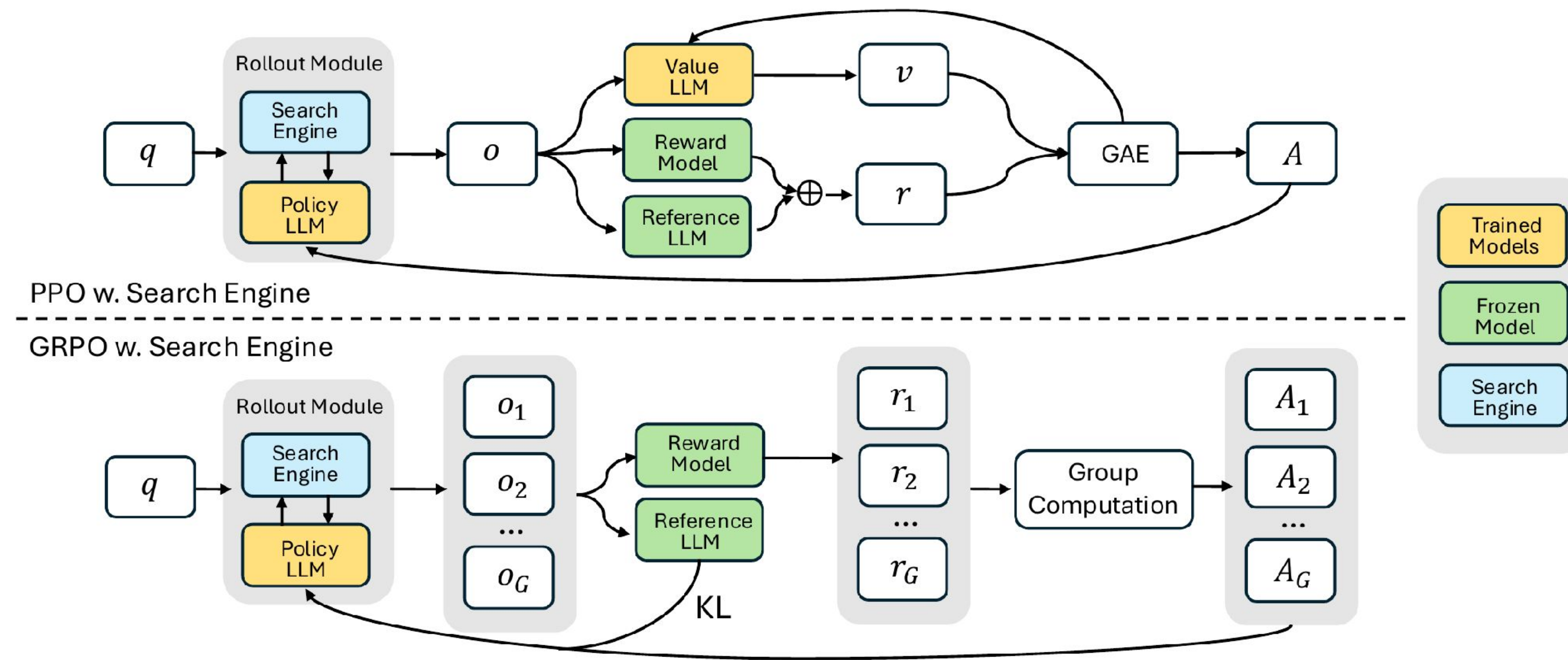
Multi-hop transfer

Train on MuSiQue

Evaluate four datasets

Search-R1

Search interaction is part of the RL rollout



PPO / GRPO

PPO is the default in the main study.

Outcome reward

Final-answer EM

Evidence Is Context, Not an Action

Retrieved-token masking

Model tokens

Think · query

m = 1

Tool tokens

Retrieved evidence

m = 0

Model tokens

Think · answer

m = 1

$$\mathcal{L}_{\text{policy}} = \frac{1}{\sum_t m_t} \sum_t m_t \ell_t^{\text{RL}}$$

The model reads the evidence; it does not learn to emit it.

Evidence: Learned Search Helps

Search-R1 · Qwen2.5-7B · Average EM across seven datasets



A stronger result in this setup—not a universal ranking.



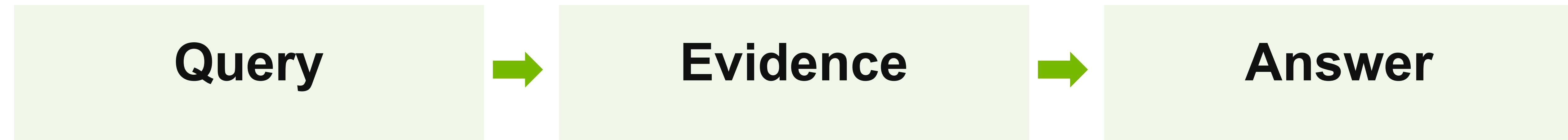
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Learning Signals: Outcome or Process?

Where Does Feedback Enter?

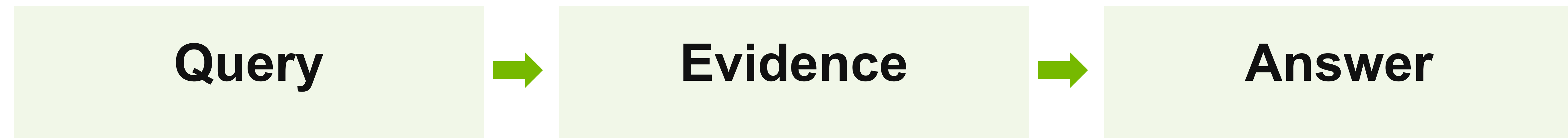
Final correctness versus intermediate decisions

Outcome



Final reward

Process



Step preference

Step preference

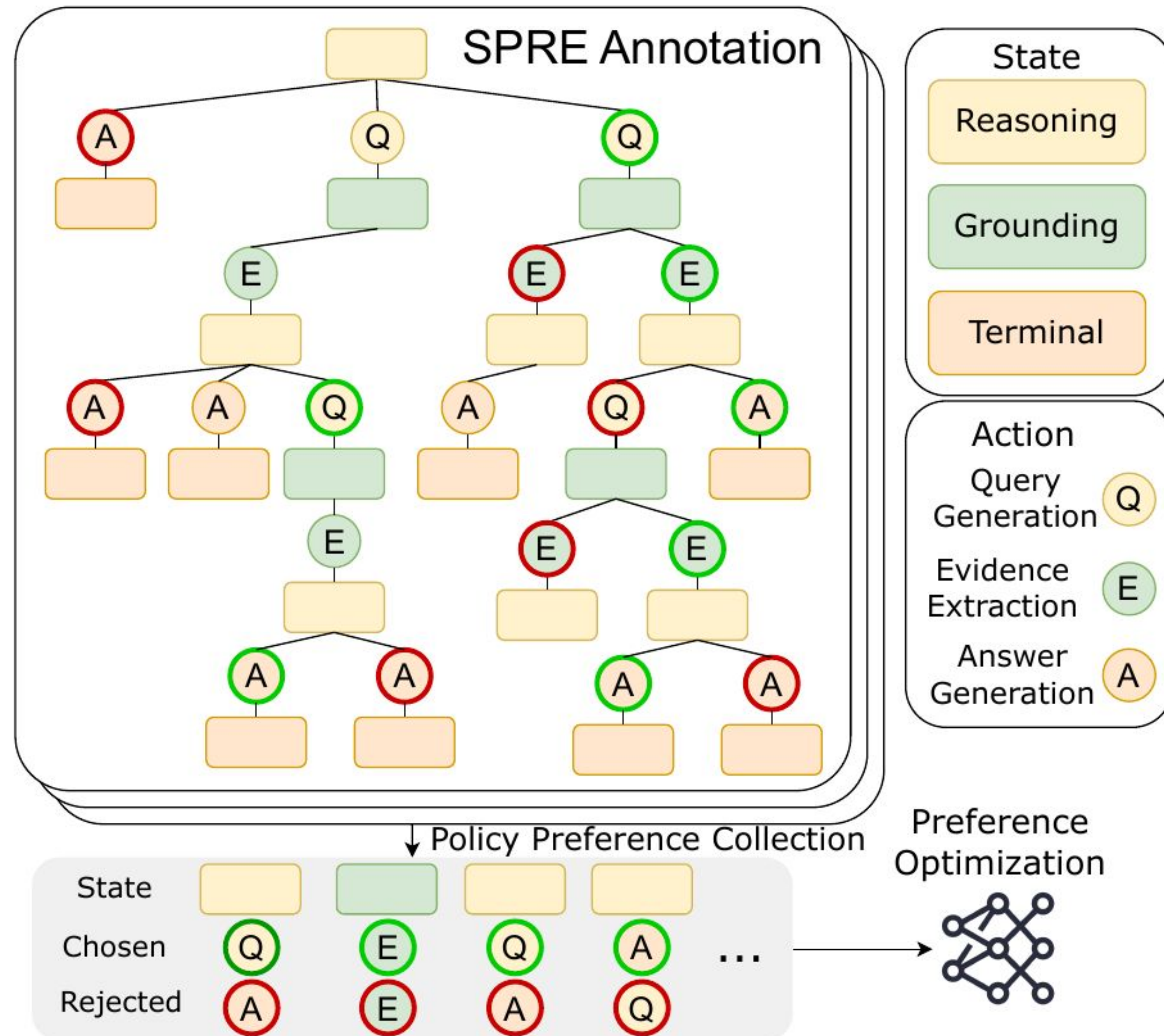
Step preference

Which earlier decision made the final answer better?

ReasonRAG

Estimate useful steps, then learn to prefer them

(a) Process-Supervised Data Generation



1 Explore

MCTS over candidate actions

2 Score

SPRE: correctness + path length

3 Optimize

DPO on step-level preferences

SPRE: Prefer Short Successful Paths

A partial state is scored through possible completions

$$Q_t = \frac{1}{k} \sum_{i=1}^k v(\tau_i) \alpha^{\text{steps}(\tau_i)}$$

v(τ): correctness

α : length discount

k: continuations

Answer F1

$0 < \alpha \leq 1$

Sampled completions

$$1 \times 0.9^3 = 0.729 > 1 \times 0.9^6 = 0.531$$

Illustration: $\alpha = 0.9$, both answers correct; 3 steps versus 6 steps

Process Supervision: Fewer Training Questions

ReasonRAG · Qwen2.5-7B-Instruct · Five QA benchmarks

Method	Train queries	Avg. EM	Avg. F1
Search-R1	90K	32.8	40.7
ReasonRAG	~5K	34.4	42.3

4,603 filtered questions → 13,289 process preference pairs

Fewer training questions ≠ proportionally less total computation

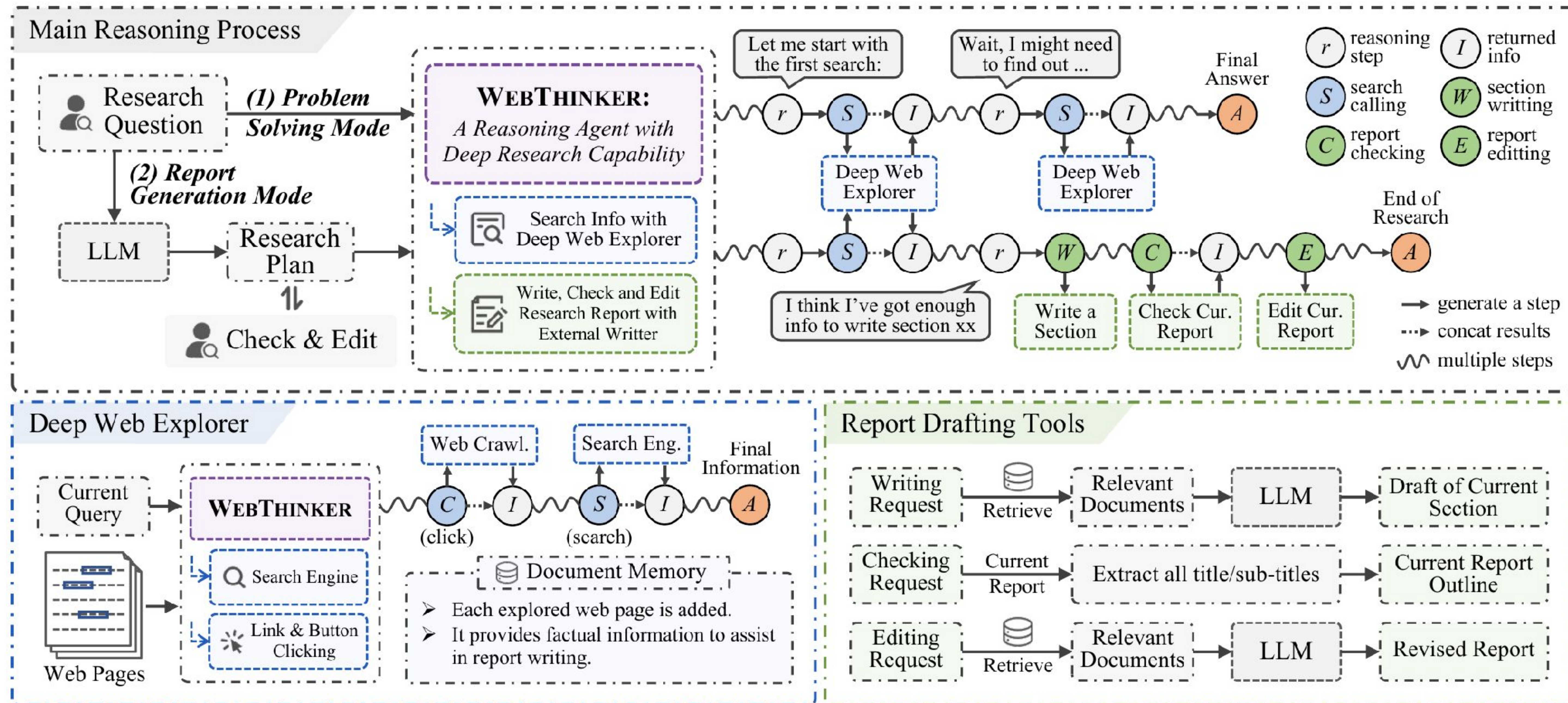


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From Search to Deep Research

WebThinker

Think, explore the web, and draft while reasoning



Search results → page exploration → evidence → answer or report

WebThinker: Iterative Online DPO

Resample trajectories as the policy improves

1

Correct / higher quality

First priority

2

Fewer tool calls

If both correct

3

Shorter reasoning

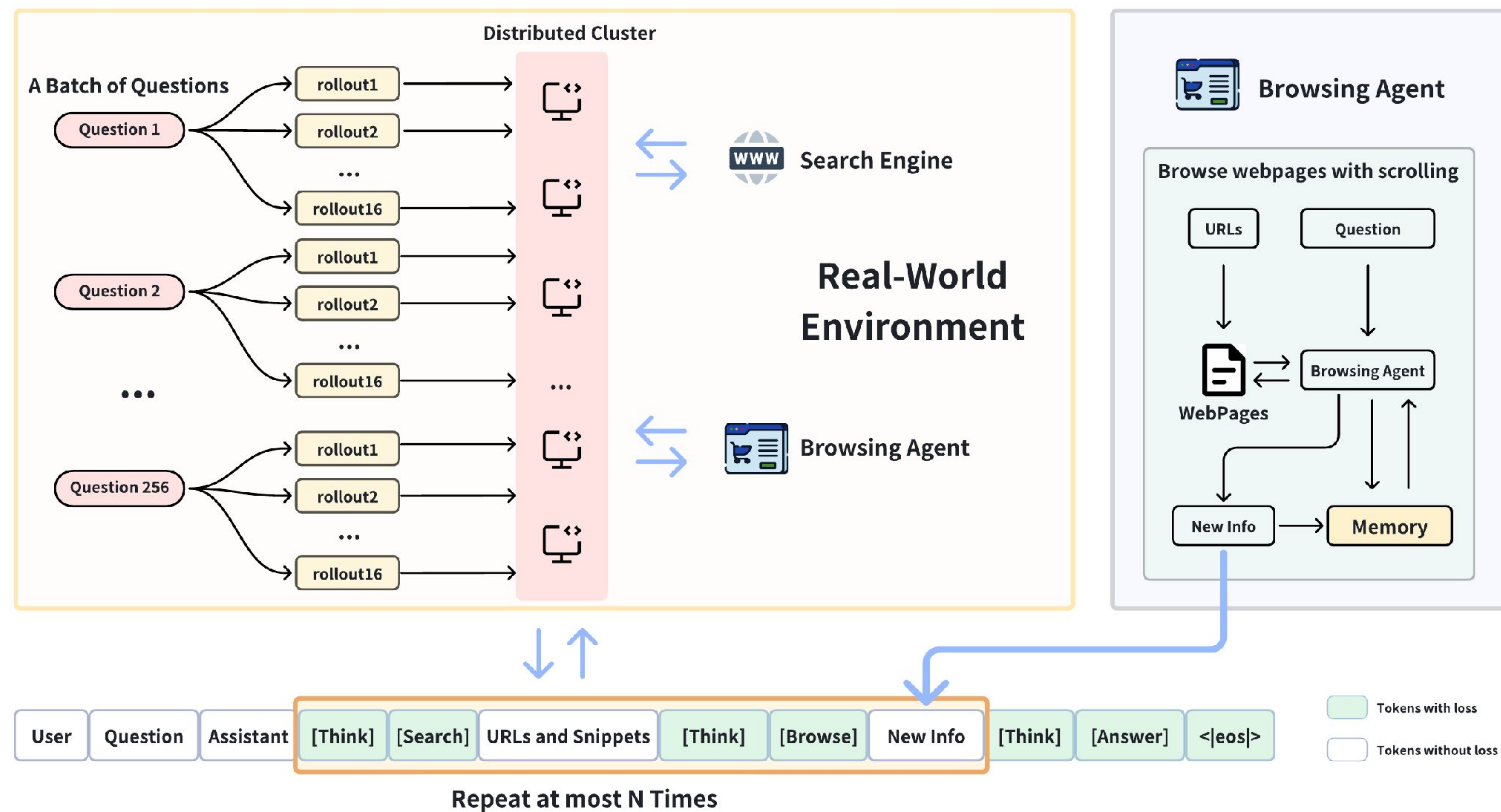
If tool use ties

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(\tau^+, \tau^-)} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(\tau^+ | q)}{\pi_{\text{ref}}(\tau^+ | q)} - \beta \log \frac{\pi_{\theta}(\tau^- | q)}{\pi_{\text{ref}}(\tau^- | q)} \right) \right]$$

Sample → rank → DPO update → sample again

DeepResearcher

Train the agent in the environment it must navigate



Real web

Search + browse
Noisy pages

GRPO

Answer F1
Format penalty

Two Meanings of “Deep Research”

Richer capabilities and more realistic training are different axes

WebThinker

Expand the toolkit

Explore · write · check · edit

QA + research reports

DeepResearcher

Change the training environment

Search · browse · verify

Primarily short-answer QA

What Does the Evidence Support?

Two within-paper comparisons—not a shared leaderboard

WebThinker · GAIA

Pass@1 (%) · QwQ-32B



+3.8 points with training

DeepResearcher · Bamboogle

Word-F1 (%) · Web-search methods



Five Papers, Three Design Choices

A synthesis of the selected 2025 research directions

Paper	Optimization	Feedback	Environment
ReSearch	GRPO	F1 + format	Wikipedia
Search-R1	PPO / GRPO	Final-answer EM	Wikipedia
ReasonRAG	MCTS → DPO	Step preferences	Wikipedia
WebThinker	Online DPO	Quality + efficiency	Open web
DeepResearcher	GRPO	F1 + format	Open web

Same broad goal. Different supervision and environments.

Evaluation Must Follow the Agent

Answer accuracy alone misses part of the system

Answer & evidence

**Correct?
Supported?**

EM / F1 · grounding

Search process

Useful next action?

Queries · evidence use

Cost & robustness

Worth another call?

Latency · failures · drift

Inspect what the agent found, used, and spent—not only its answer.

The Research Shift

From retrieving documents to learning how to acquire evidence

Control

Search is an action

Not only a preprocessing
step

Credit assignment

**Feedback shapes
behavior**

Not only the final answer

Environment

**Training must meet
reality**

Not only a clean corpus

Learn when to search, what to inspect, and when to stop.



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Appendix

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Appendix A1: Search-R1 Results

Qwen2.5-7B block

Methods	General QA				Multi-Hop QA			
	NQ [†]	TriviaQA [*]	PopQA [*]	HotpotQA [†]	2wiki [*]	Musique [*]	Bamboogle [*]	Avg.
Qwen2.5-7b-Base/Instruct								
Direct Inference	0.134	0.408	0.140	0.183	0.250	0.031	0.120	0.181
CoT	0.048	0.185	0.054	0.092	0.111	0.022	0.232	0.106
IRCoT	0.224	0.478	0.301	0.133	0.149	0.072	0.224	0.239
Search-o1	0.151	0.443	0.131	0.187	0.176	0.058	0.296	0.206
RAG	0.349	0.585	0.392	0.299	0.235	0.058	0.208	0.304
SFT	0.318	0.354	0.121	0.217	0.259	0.066	0.112	0.207
R1-base	0.297	0.539	0.202	0.242	0.273	0.083	0.296	0.276
R1-instruct	0.270	0.537	0.199	0.237	0.292	0.072	0.293	0.271
Rejection Sampling	0.360	0.592	0.380	0.331	0.296	0.123	0.355	0.348
Search-R1-base	0.480	0.638	0.457	0.433	0.382	0.196	0.432	0.431
Search-R1-instruct	0.393	0.610	0.397	0.370	0.414	0.146	0.368	0.385

Appendix A2: ReasonRAG Results

Five QA benchmarks

Type	Method	PopQA		HotpotQA		2WikiMulti		Bamboogle		MuSiQue		Avg.	
		EM	F ₁	EM	F ₁	EM	F ₁	EM	F ₁	EM	F ₁	EM	F ₁
Zero-shot	Naïve Generation	12.7	16.5	15.7	24.8	20.2	28.0	6.4	17.4	2.7	10.2	11.5	19.4
	Standard RAG	38.4	44.7	29.3	39.9	29.4	36.3	17.6	24.1	6.7	15.1	24.3	32.0
Active	FLARE	14.3	17.6	18.1	25.7	27.9	32.8	12.0	20.8	4.3	12.6	15.3	21.9
	Self-RAG(146k)	22.7	33.9	21.0	29.7	12.0	25.2	1.6	10.9	4.6	13.3	12.4	22.6
Adaptive	AdaptiveRAG(3k)	36.6	41.5	29.1	40.7	24.2	33.4	18.4	26.1	6.9	14.3	23.0	31.2
RAG-CoT	Iter-Retgen	38.7	44.9	30.3	42.1	31.2	38.7	19.2	26.4	7.7	14.2	25.4	33.3
	IRCoT	36.2	43.6	27.7	41.5	23.5	32.5	17.2	22.5	8.6	13.2	22.6	30.7
Summary	RECOMP	<u>40.5</u>	<u>45.8</u>	29.7	41.2	33.2	39.4	21.7	28.6	9.2	15.8	26.9	34.2
	LongLLMLingua	39.2	45.1	31.4	43.2	34.5	40.2	20.3	27.4	8.7	14.9	26.8	34.2
	Selective-Context	34.9	41.5	19.3	27.3	20.3	29.7	15.3	22.6	6.1	13.7	19.2	27.0
Reasoning	Search-o1	33.2	40.3	24.8	38.1	16.4	27.1	30.4	40.6	6.3	13.7	22.2	31.96
	AutoRAG(10k)	38.6	44.1	33.3	43.7	39.5	46.1	24.8	32.2	11.3	18.3	29.5	36.9
	Search-R1(90k)	39.7	44.8	<u>37.0</u>	<u>47.0</u>	<u>41.4</u>	<u>48.0</u>	<u>32.0</u>	<u>43.8</u>	14.6	<u>19.9</u>	<u>32.8</u>	<u>40.7</u>
	ReasonRAG(5k)	41.5*	46.2*	38.4*	48.9*	43.6*	50.4*	36.0*	45.5*	<u>12.8</u>	20.6*	34.4*	42.3*

Appendix A3: DeepResearcher Results

Out-of-domain evaluation

Method	Inference Environment	Musique		Bamboogle		PopQA	
		F1	MBE	F1	MBE	F1	MBE
<i>Prompt Based</i>							
CoT	Local RAG	8.5	7.4	22.1	21.6	17.0	15.0
CoT + RAG	Local RAG	10.0	10.0	25.4	27.2	46.9	48.8
Search-o1*	Local RAG	16.8	21.3	35.8	38.4	36.9	42.4
Search-o1	Web Search	14.7	19.7	46.6	53.6	38.3	43.4
ReAct-style Agent	Web Search	8.9	10.0	34.4	36.8	19.1	20.5
<i>Training Based</i>							
Search-r1-base	Local RAG	26.7	27.5	56.5	57.6	43.2	47.0
Search-r1-instruct	Local RAG	26.5	28.3	45.0	47.2	43.0	44.5
R1-Searcher	Web Search	22.8	25.6	64.8	65.6	42.7	43.4
DeepResearcher (Local RAG)	Local RAG	12.7	12.5	42.7	46.4	23.2	23.4
DeepResearcher	Web Search	27.1	29.3	71.0	72.8	48.5	52.7



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